

to deride symbolic approaches, not all philosophers are critical of GOFAI. Some are, in fact, ardent advocates and even practitioners. Zenon Pylyshyn (1984) has argued that cognition can best be understood through a computational model, not only in principle but also as a way of conducting research at present, and has specifically rebutted Dreyfus's criticisms of the computational model of human cognition (Pylyshyn, 1974). Gilbert Harman (1983), in analyzing belief revision, makes connections with AI research on truth maintenance systems. Michael Bratman has applied his "belief-desire-intention" model of human psychology (Bratman, 1987) to AI research on planning (Bratman, 1992). At the extreme end of strong AI, Aaron Sloman (1978, p. xiii) has even described as "racialist" the claim by Joseph Weizenbaum (1976) that intelligent machines can never be regarded as persons.

Proponents of the importance of embodiment in cognition include the philosophers Merleau-Ponty, whose *Phenomenology of Perception* (1945) stressed the importance of the body and the subjective interpretation of reality afforded by our senses, and Heidegger, whose *Being and Time* (1927) asked what it means to actually be an agent, and criticized all of the history of philosophy for taking this notion for granted. In the computer age, Alva Noe (2009) and Andy Clark (1998, 2008) propose that our brains form a rather minimal representation of the world, use the world itself in a just-in-time basis to maintain the illusion of a detailed internal model, use props in the world (such as paper and pencil as well as computers) to increase the capabilities of the mind. Pfeifer *et al.* (2006) and Lakoff and Johnson (1999) present arguments for how the body helps shape cognition.

The nature of the mind has been a standard topic of philosophical theorizing from ancient times to the present. In the *Phaedo*, Plato specifically considered and rejected the idea that the mind could be an "attunement" or pattern of organization of the parts of the body, a viewpoint that approximates the functionalist viewpoint in modern philosophy of mind. He decided instead that the mind had to be an immortal, immaterial soul, separable from the body and different in substance—the viewpoint of dualism. Aristotle distinguished a variety of souls (Greek $\psi\upsilon\chi\eta$) in living things, some of which, at least, he described in a functionalist manner. (See Nussbaum (1978) for more on Aristotle's functionalism.)

Descartes is notorious for his dualistic view of the human mind, but ironically his historical influence was toward mechanism and physicalism. He explicitly conceived of animals as automata, and he anticipated the Turing Test, writing "it is not conceivable [that a machine] should produce different arrangements of words so as to give an appropriately meaningful answer to whatever is said in its presence, as even the dullest of men can do" (Descartes, 1637). Descartes's spirited defense of the animals-as-automata viewpoint actually had the effect of making it easier to conceive of humans as automata as well, even though he himself did not take this step. The book *L'Homme Machine* (La Mettrie, 1748) did explicitly argue that humans are automata.

Modern analytic philosophy has typically accepted physicalism, but the variety of views on the content of mental states is bewildering. The identification of mental states with brain states is usually attributed to Place (1956) and Smart (1959). The debate between narrow-content and wide-content views of mental states was triggered by Hilary Putnam (1975), who introduced so-called **twin earths** (rather than brain-in-a-vat, as we did in the chapter) as a device to generate identical brain states with different (wide) content.

Functionalism is the philosophy of mind most naturally suggested by AI. The idea that mental states correspond to classes of brain states defined functionally is due to Putnam (1960, 1967) and Lewis (1966, 1980). Perhaps the most forceful proponent of functionalism is Daniel Dennett, whose ambitiously titled work *Consciousness Explained* (Dennett, 1991) has attracted many attempted rebuttals. Metzinger (2009) argues there is no such thing as an objective *self*, that consciousness is the subjective appearance of a world. The inverted spectrum argument concerning qualia was introduced by John Locke (1690). Frank Jackson (1982) designed an influential thought experiment involving Mary, a color scientist who has been brought up in an entirely black-and-white world. *There's Something About Mary* (Ludlow *et al.*, 2004) collects several papers on this topic.

Functionalism has come under attack from authors who claim that they do not account for the *qualia* or “what it’s like” aspect of mental states (Nagel, 1974). Searle has focused instead on the alleged inability of functionalism to account for intentionality (Searle, 1980, 1984, 1992). Churchland and Churchland (1982) rebut both these types of criticism. The Chinese Room has been debated endlessly (Searle, 1980, 1990; Preston and Bishop, 2002). We’ll just mention here a related work: Terry Bisson’s (1990) science fiction story *They’re Made out of Meat*, in which alien robotic explorers who visit earth are incredulous to find thinking human beings whose minds are made of meat. Presumably, the robotic alien equivalent of Searle believes that he can think due to the special causal powers of robotic circuits; causal powers that mere meat-brains do not possess.

Ethical issues in AI predate the existence of the field itself. I. J. Good’s (1965) ultraintelligent machine idea was foreseen a hundred years earlier by Samuel Butler (1863). Written four years after the publication of Darwin’s *On the Origins of Species* and at a time when the most sophisticated machines were steam engines, Butler’s article on *Darwin Among the Machines* envisioned “the ultimate development of mechanical consciousness” by natural selection. The theme was reiterated by George Dyson (1998) in a book of the same title.

The philosophical literature on minds, brains, and related topics is large and difficult to read without training in the terminology and methods of argument employed. The *Encyclopedia of Philosophy* (Edwards, 1967) is an impressively authoritative and very useful aid in this process. *The Cambridge Dictionary of Philosophy* (Audi, 1999) is a shorter and more accessible work, and the online *Stanford Encyclopedia of Philosophy* offers many excellent articles and up-to-date references. The *MIT Encyclopedia of Cognitive Science* (Wilson and Keil, 1999) covers the philosophy of mind as well as the biology and psychology of mind. There are several general introductions to the philosophical “AI question” (Boden, 1990; Haugeland, 1985; Copeland, 1993; McCorduck, 2004; Minsky, 2007). *The Behavioral and Brain Sciences*, abbreviated *BBS*, is a major journal devoted to philosophical and scientific debates about AI and neuroscience. Topics of ethics and responsibility in AI are covered in the journals *AI and Society* and *Journal of Artificial Intelligence and Law*.

EXERCISES

26.1 Go through Turing's list of alleged "disabilities" of machines, identifying which have been achieved, which are achievable in principle by a program, and which are still problematic because they require conscious mental states.

26.2 Find and analyze an account in the popular media of one or more of the arguments to the effect that AI is impossible.

26.3 In the brain replacement argument, it is important to be able to restore the subject's brain to normal, such that its external behavior is as it would have been if the operation had not taken place. Can the skeptic reasonably object that this would require updating those neurophysiological properties of the neurons relating to conscious experience, as distinct from those involved in the functional behavior of the neurons?

26.4 Suppose that a Prolog program containing many clauses about the rules of British citizenship is compiled and run on an ordinary computer. Analyze the "brain states" of the computer under wide and narrow content.

26.5 Alan Perlis (1982) wrote, "A year spent in artificial intelligence is enough to make one believe in God". He also wrote, in a letter to Philip Davis, that one of the central dreams of computer science is that "through the performance of computers and their programs we will remove all doubt that there is only a chemical distinction between the living and nonliving world." To what extent does the progress made so far in artificial intelligence shed light on these issues? Suppose that at some future date, the AI endeavor has been completely successful; that is, we have build intelligent agents capable of carrying out any human cognitive task at human levels of ability. To what extent would that shed light on these issues?

26.6 Compare the social impact of artificial intelligence in the last fifty years with the social impact of the introduction of electric appliances and the internal combustion engine in the fifty years between 1890 and 1940.

26.7 I. J. Good claims that intelligence is the most important quality, and that building ultraintelligent machines will change everything. A sentient cheetah counters that "Actually speed is more important; if we could build ultrafast machines, that would change everything," and a sentient elephant claims "You're both wrong; what we need is ultrastrong machines." What do you think of these arguments?

26.8 Analyze the potential threats from AI technology to society. What threats are most serious, and how might they be combated? How do they compare to the potential benefits?

26.9 How do the potential threats from AI technology compare with those from other computer science technologies, and to bio-, nano-, and nuclear technologies?

26.10 Some critics object that AI is impossible, while others object that it is *too* possible and that ultraintelligent machines pose a threat. Which of these objections do you think is more likely? Would it be a contradiction for someone to hold both positions?